



DPUNav

Simulation of a simple navigation system.

1. Inputs:

<i>Show</i>	=true: navigation system is visible for the driver =false: navigation system is not visible for the driver
<i>Image</i>	Selection of the displayed image.

2. Dynamic parameters:

BackgroundR, BackgroundG, BackgroundB

Background color of the navigation window. Each value's range is from 0 (dark) to 255 (bright).

BackgroundB

The default value for each variable is 0 (black).

EraseBackground

=true (default): Erase the background before drawing another image

=false: Draw the images on top of each other

Scale

=true (default): Scale the images to fit in the window. The aspect ratio of the images is maintained.

=false: Draw the images at their original size. They are centered within the window.

Topmost

=true: the window for the navigation system will stay on top of all other windows.

=false (default): Other windows can be drawn on top of the navigation window.

3. Static parameters:

x, y

Window position of the navigation system. The coordinates are specified in pixels and the origin is in the upper left corner.

Width, Height

Width and Height of the window.

BitmapPath

Path of the images for navigation.

Default: `\SILAB\lib\graphics\default\nav`

Bitmap[1...20]

Images that are shown for navigation. They all have to be located in the BitmapPath and saved in windows bitmap format (BMP).

As default setting all Bitmap parameters are related to certain navigation arrows (see example below).

4. Functional principle:

DPUNav shows a navigation image depending on the input value Image. The selection of images that can be displayed are defined via parameter Bitmap.

For the value *Image*=0 the Bitmap1 is displayed, for the value *Image*=1 the Bitmap2 is displayed etc.



5. Example:

For using the standard navigation arrows no parameters have to be specified. In this case the values of the parameter Bitmap are as follows:

Image	BMP file name	Variable
0	Clear	Empty window
1	Straight	Arrow „straight ahead“
2	Left	Arrow „turn left“
3	Right	Arrow „turn right“
4	TurnRound	Arrow „turn-over“
5	InsideLeft	Arrow „slight turn left“
6	InsideRight	Arrow „slight turn right“

DPUNav is often controlled by hedgehogs. To do this, a connection to an instance of DPUSCNXHedgehogKiller is necessary. The configuration below shows an ex-ample:

```
DPUSCNXHedgehogKiller NavHedgehogKiller
{
    Computer = {VDYN};
    Index = 100;

    Family = "Nav";
    Name1 = "Image";
    Default1 = 0.0;
};

DPUNav Nav
{
    Computer = {DSP};
    Index = 50;
};

Connections = {
    NavHedgehogKiller.Out1 <-> Nav.Image
};
```

Appropriate hedgehogs can be defined as follows:

```
Hedgehogs =
{
    # (Course, position along road, Direction, LaneID, Family, Name1,
    Value)
    (Course, 100, DontCare, 0, Nav, Image, 1) # straight ahead
};
```